

Spindle / Main Bearing Replacement

Standard Operating Procedure

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Version	2.0
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Owner	Maintenance Department
Scope	CNC Milling units (all plants); applicable to spindle and drive-shaft main bearings

1. Purpose

Defines the safe, standardized method for diagnosing and replacing a failed spindle or main bearing following a DT-08 fault, to restore machine availability and prevent repeat failure.

2. Required PPE

- Safety glasses
- Cut-resistant gloves
- Steel-toe boots

3. Tools & Materials

- Bearing puller kit
- Torque wrench (calibrated)
- Induction bearing heater
- Replacement bearing (matched to machine spec plate)
- Dial indicator

4. Procedure

- 1 Lockout-tagout the machine per Safety_Guidelines before any disassembly.
- 2 Confirm DT-08 fault via controller alarm log and cross-check spindle vibration/noise symptoms.
- 3 Remove spindle guard and drain surrounding lubricant reservoir.
- 4 Use bearing puller to remove the failed bearing; inspect shaft journal for scoring.
- 5 Induction-heat the replacement bearing to 80–90°C for interference fit; press onto shaft.
- 6 Re-torque housing bolts to spec (see machine manual) in a star pattern.
- 7 Re-lubricate per Lubrication_Schedule and rotate shaft by hand to confirm free movement.
- 8 Run machine at low RPM for 10 minutes, monitor vibration and temperature, then return to production.
- 9 Log parts used, downtime duration, and technician ID in the maintenance CMMS record.

5. Associated Downtime Code

Code	Description	Severity	Typical Duration
DT-08	Spindle / Main Bearing Failure	Major	57 min

■ **Never use an open flame to heat a bearing. Use only an approved induction heater to avoid metallurgical damage.**